10000000000000000000000000000000000000000 (that's 1 followed by 43 zeros). Suffice it to say, this is an unimaginably small amount of time. And, to be clear, it's not that we necessarily *can* explain everything from the Planck Time on, but that we currently definitely *cannot* explain anything before it.

To sum up where we've gotten so far: there may have been a singularity. If there was, it was immediately followed by an era we can't really say much about called the Planck Time.

Truthfully, the whole timeline of the early universe is still very much an extrapolation and, I will readily admit, one that we shouldn't entirely trust. A universe that starts with a singularity and expands from there goes through an unimaginably extreme range of temperatures, from basically infinity at the singularity to the cool comfortable environment of the cosmos today, sitting at about 3 degrees above absolute zero. What we can do is make inferences about what physics would be like in all those environments, which is how we get the ordering I present in this chapter. And though the standard Big Bang theory of steady expansion from a singularity has some major problems (which we'll get to imminently), we can still learn a lot about how physics works by thinking about what might have happened if the standard theory is right.

THE GUT ERA

According to the standard Big Bang story, after the Planck Time comes the GUT era. (I'm using the term "era" here to refer to something that lasts about 10^{-35} seconds, and the term GUT to mean something unrelated to human anatomy.) "GUT" stands for Grand Unified Theory, which is the physics-utopian ideal of a "unified" theory that describes how all the forces in particle physics worked together under the extreme conditions of the universe at this early stage. Even though the universe was cooling rapidly, it was still so hot that the amount of energy at every point in space was over a trillion times higher than the energy generated by the most powerful collisions in our most advanced particle colliders. Unfortunately, and partly because of that factor of a trillion preventing us from doing experimental tests, the theory is currently still very much under construction. But for a theory we do not currently have, we can say quite a lot about it, and how it's different from what we see today.